

Homework 02

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Problem 1. Let X , Y and Z be random variables and let c be a number. Prove that

$$\text{Var}(cX) = c^2 \text{Var}(X),$$

$$\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$$

and

$$\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z).$$

(i) $\text{Var}(cX) = c^2 \text{Var}(X)$.

$$\begin{aligned} \text{Var}(cX) &= \mathbb{E}\left((cX - \mathbb{E}(cX))^2\right) \\ &= \mathbb{E}\left(c^2 X^2 - 2cX \mathbb{E}(cX) + (\mathbb{E}(cX))^2\right) \\ &= \mathbb{E}\left(c^2 X^2 - 2c^2 X \mathbb{E}(X) + (c \mathbb{E}(X))^2\right) \\ &= \mathbb{E}\left(c^2 X^2 - 2c^2 X \mathbb{E}(X) + c^2 (\mathbb{E}(X))^2\right) \\ &= \mathbb{E}\left(c^2 \left(X^2 - 2X \mathbb{E}(X) + (\mathbb{E}(X))^2\right)\right) \\ &= c^2 \mathbb{E}\left(X^2 - 2X \mathbb{E}(X) + (\mathbb{E}(X))^2\right) \\ &= c^2 \mathbb{E}\left((X - \mathbb{E}(X))^2\right) \\ &= c^2 \text{Var}(X). \end{aligned}$$

(ii) $\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$. Starting from the deviation definition $\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$, and using $\mu_{cX} = c \mu_X$:

$$\begin{aligned} \text{Cov}(cX, Y) &= \mathbb{E}[(cX - \mu_{cX})(Y - \mu_Y)] \\ &= \mathbb{E}[(cX - c\mu_X)(Y - \mu_Y)] \\ &= \mathbb{E}[c(X - \mu_X)(Y - \mu_Y)] \\ &= c \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ &= c \text{Cov}(X, Y). \end{aligned}$$

(iii) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$.

$$\begin{aligned}
 \text{Cov}(X + Y, Z) &= \mathbb{E}[(X + Y - \mu_{X+Y})(Z - \mu_Z)] \\
 &= \mathbb{E}[(X + Y - \mu_X - \mu_Y)(Z - \mu_Z)] \\
 &= \mathbb{E}[(X - \mu_X + Y - \mu_Y)(Z - \mu_Z)] \\
 &= \mathbb{E}[(X - \mu_X)(Z - \mu_Z) + (Y - \mu_Y)(Z - \mu_Z)] \\
 &= \mathbb{E}[(X - \mu_X)(Z - \mu_Z)] + \mathbb{E}[(Y - \mu_Y)(Z - \mu_Z)] \\
 &= \text{Cov}(X, Z) + \text{Cov}(Y, Z).
 \end{aligned}$$

The collapse $\mu_{X+Y} = \mu_X + \mu_Y$ on the second line, and the split of the expectation on the fifth, are both linearity of expectation — which is exactly what Problem 2 proves.

Problem 2. Let X_1 and X_2 be discrete random variables and suppose that a_1 and a_2 are real numbers. Using the joint probability mass function for X_1 and X_2 , show that

$$\mathbb{E}[a_1X_1 + a_2X_2] = a_1\mathbb{E}[X_1] + a_2\mathbb{E}[X_2].$$

Write $p(x_1, x_2)$ for the joint probability mass function of X_1 and X_2 , and p_{X_1} , p_{X_2} for the marginals.

$$\begin{aligned}
 \mathbb{E}[a_1X_1 + a_2X_2] &= \sum_{x_1} \sum_{x_2} (a_1x_1 + a_2x_2) p(x_1, x_2) \\
 &= \sum_{x_1} \sum_{x_2} a_1x_1 p(x_1, x_2) + \sum_{x_1} \sum_{x_2} a_2x_2 p(x_1, x_2) \\
 &= \sum_{x_1} a_1x_1 p_{X_1}(x_1) + \sum_{x_2} \sum_{x_1} a_2x_2 p(x_1, x_2) \\
 &= \sum_{x_1} a_1x_1 p_{X_1}(x_1) + \sum_{x_2} a_2x_2 p_{X_2}(x_2) \\
 &= a_1 \sum_{x_1} x_1 p_{X_1}(x_1) + a_2 \sum_{x_2} x_2 p_{X_2}(x_2) \\
 &= a_1\mathbb{E}[X_1] + a_2\mathbb{E}[X_2].
 \end{aligned}$$

The third and fourth lines are marginalization: summing the joint pmf over the variable that does not appear in the summand leaves the marginal pmf of the other.

Problem 9. (Edition 11, Chapter 2.) If the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0, \\ \frac{1}{2} & 0 \leq b < 1, \\ \frac{3}{5} & 1 \leq b < 2, \\ \frac{4}{5} & 2 \leq b < 3, \\ \frac{9}{10} & 3 \leq b < 3.5, \\ 1 & b \geq 3.5, \end{cases}$$

calculate the probability mass function of X .

The mass at each point is the size of the jump of F there, so subtract consecutive values of F : $p(1) = F(1) - F(0)$, and so on.

$$p(0) = \frac{1}{2}, \quad p(1) = \frac{1}{10}, \quad p(2) = \frac{1}{5}, \quad p(3) = \frac{1}{10}, \quad p(3.5) = \frac{1}{10},$$

and $p(x) = 0$ for every other x .

Problem 12. (Edition 11, Chapter 2.) On a multiple-choice exam with three possible answers for each of the five questions, what is the probability that a student would get four or more correct answers just by guessing?

Five questions, three options each, so the number correct is Binomial(5, 1/3) and $\mathbb{P}(\text{correct}) = \frac{1}{3}$ on any one question. Four or more correct means four or five:

$$\begin{aligned}\mathbb{P}(4) + \mathbb{P}(5) &= \binom{5}{4} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)^5 \\ &= \frac{10}{243} + \frac{1}{243} = \boxed{\frac{11}{243}}.\end{aligned}$$

Problem 22. (Edition 11, Chapter 2.) If a fair coin is successively flipped, find the probability that a head first appears on the fifth trial.

A first head on the fifth trial is exactly the one sequence $TTTTH$, and the flips are independent and fair, so

$$\mathbb{P}(\text{first head on trial 5}) = \left(\frac{1}{2}\right)^5 = \frac{1}{32}.$$

Problem 37. (Edition 11, Chapter 2.) Let $X_1, X_2, \dots, X_n$ be independent random variables, each having a uniform distribution over $(0, 1)$. Let $M = \text{maximum}(X_1, X_2, \dots, X_n)$. Show that the distribution function of M , $F_M(\cdot)$, is given by

$$F_M(x) = x^n, \quad 0 \leq x \leq 1.$$

What is the probability density function of M ?

The maximum is at most x exactly when every X_i is at most x . Concretely, for $n = 3$ and $x = \frac{1}{2}$, the event is $X_1 \leq \frac{1}{2}, X_2 \leq \frac{1}{2}, X_3 \leq \frac{1}{2}$, which by independence has probability $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^3$. In general, using independence and $\mathbb{P}(X_i \leq x) = x$ for $0 \leq x \leq 1$,

$$F_M(x) = \mathbb{P}(M \leq x) = \mathbb{P}(X_1 \leq x) \mathbb{P}(X_2 \leq x) \cdots \mathbb{P}(X_n \leq x) = x^n, \quad 0 \leq x \leq 1.$$

The curve runs from 0 to 1 and is steeper near 1 as n grows. Differentiating,

$$f_M(x) = \frac{d}{dx} [x^n] = nx^{n-1}, \quad 0 \leq x \leq 1.$$

Problem 61. (Edition 11, Chapter 2.) Let X and W be the working and subsequent repair times of a certain machine. Let $Y = X + W$ and suppose that the joint probability density of X and Y is

$$f_{X,Y}(x, y) = \lambda^2 e^{-\lambda y}, \quad 0 < x < y < \infty.$$

1. Find the density of X .
2. Find the density of Y .
3. Find the joint density of X and W .
4. Find the density of W .

(a) **The density of X .** Integrate the joint density over y . The region is $0 < x < y < \infty$, so for a fixed x the variable y runs from x up to ∞ :

$$f_X(x) = \int_y f_{X,Y}(x, y) dy = \int_x^\infty \lambda^2 e^{-\lambda y} dy.$$

Substituting $u = -\lambda y$, so that $du = -\lambda dy$, $dy = -du/\lambda$, and $y = x$ becomes $u = -\lambda x$:

$$\begin{aligned} f_X(x) &= \int_{-\lambda x}^{-\infty} \lambda^2 e^u \left(-\frac{du}{\lambda} \right) = \lambda \int_{-\infty}^{-\lambda x} e^u du = \lambda [e^u]_{-\infty}^{-\lambda x} \\ &= \lambda [e^{-\lambda x} - 0] = \boxed{\lambda e^{-\lambda x}}, \quad x > 0. \end{aligned}$$

(b) **The density of Y .** Integrate the joint density over x instead. For a fixed y , the region $0 < x < y$ makes x run from 0 to y , and $\lambda^2 e^{-\lambda y}$ is constant in x :

$$\begin{aligned} f_Y(y) &= \int_x f_{X,Y}(x, y) dx = \int_0^y \lambda^2 e^{-\lambda y} dx = [x \lambda^2 e^{-\lambda y}]_0^y \\ &= \boxed{\lambda^2 y e^{-\lambda y}}, \quad y \in (0, \infty). \end{aligned}$$

(c) **The joint density of X and W .** Here $Y = X + W$, so substituting $y = x + w$ into $f_{X,Y}$ turns the exponent into a sum, and the constraint $x < y$ becomes $w > 0$:

$$f_{X,W}(x, w) = \boxed{\lambda^2 e^{-(x+w)\lambda}}, \quad x > 0, w > 0.$$

(d) **The density of W .** Integrate the joint density of part (c) over x , which now runs from 0 to ∞ . Substituting $u = -(x + w)\lambda$, so $du = -\lambda dx$ and $dx = -du/\lambda$:

$$\begin{aligned} f_W(w) &= \int_0^\infty \lambda^2 e^{-(x+w)\lambda} dx = \int_{-\lambda w}^{-\infty} -\lambda e^u du = \lambda \int_{-\infty}^{-\lambda w} e^u du \\ &= \boxed{\lambda e^{-\lambda w}}, \quad w > 0. \end{aligned}$$

(e) **X and W are independent.** Multiplying the two marginals from (a) and (d) reproduces the joint density from (c):

$$f_X(x) f_W(w) = \lambda^2 e^{-\lambda x} e^{-\lambda w} = \lambda^2 e^{-(x+w)\lambda} = f_{X,W}(x, w),$$

therefore X and W are independent.

Problem 76. (*Edition 11, Chapter 2.*) Let X and Y be independent random variables with means μ_x and μ_y and variances σ_x^2 and σ_y^2 . Show that

$$\text{Var}(XY) = \sigma_x^2 \sigma_y^2 + \mu_y^2 \sigma_x^2 + \mu_x^2 \sigma_y^2.$$

Start from the computational form of the variance, applied to the single random variable XY , and use independence twice — once on $\mathbb{E}[X^2 Y^2]$ and once on $\mathbb{E}[XY]$:

$$\begin{aligned} \text{Var}[XY] &= \mathbb{E}[(XY)^2] - (\mathbb{E}[XY])^2 \\ &= \mathbb{E}[X^2 Y^2] - (\mathbb{E}[X] \mathbb{E}[Y])^2 && (X, Y \text{ independent}) \\ &= \mathbb{E}[X^2] \mathbb{E}[Y^2] - \mathbb{E}^2[X] \cdot \mathbb{E}^2[Y]. \end{aligned}$$

Now substitute the second moments, using $\mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}(X))^2 = \sigma_x^2 + \mu_x^2$ and likewise for Y :

$$\begin{aligned}\text{Var}[XY] &= (\sigma_x^2 + \mu_x^2)(\sigma_y^2 + \mu_y^2) - \mu_x^2 \cdot \mu_y^2 \\ &= \sigma_x^2 \sigma_y^2 + \mu_y^2 \sigma_x^2 + \mu_x^2 \sigma_y^2 + \mu_x^2 \mu_y^2 - \mu_x^2 \mu_y^2 \\ &= \sigma_x^2 \sigma_y^2 + \mu_y^2 \sigma_x^2 + \mu_x^2 \sigma_y^2. \quad \therefore\end{aligned}$$

The $+\mu_x^2 \mu_y^2$ produced by expanding the product cancels the $-\mu_x^2 \mu_y^2$ carried down from $(\mathbb{E}[XY])^2$, leaving the required three terms.

Problem 86. (From Edition 11.) Each new book donated to a library must be processed. Suppose that the time it takes a librarian to process a book has mean 10 minutes and standard deviation 3 minutes. If a librarian has 40 books that must be processed one at a time,

1. approximate the probability that it will take more than 420 minutes to process all these books;
2. approximate the probability that at least 25 books will be processed in the first 240 minutes.

Let T_i be the time taken to process book i . The T_i are independent and identically distributed with $\mathbb{E}[T_i] = 10$ minutes and $\text{SD}(T_i) = 3$ minutes, so $\text{Var}(T_i) = 9 \text{ min}^2$. Because the books are processed one at a time, the moment book n is finished is

$$S_n = T_1 + T_2 + \cdots + T_n,$$

and by linearity of expectation and independence,

$$\mathbb{E}[S_n] = 10n, \quad \text{Var}(S_n) = 9n, \quad \text{SD}(S_n) = 3\sqrt{n}.$$

The standard deviation is the square root of the variance of the sum; it is not n times $\text{SD}(T_i)$. By the central limit theorem, for n this large

$$Z = \frac{S_n - \mathbb{E}[S_n]}{\text{SD}(S_n)}$$

is approximately standard normal, and $\Phi(z) = \mathbb{P}(Z \leq z)$ denotes the standard normal distribution function.

(a) More than 420 minutes to process all 40 books. All 40 books are finished at time S_{40} , with

$$\begin{aligned}\mathbb{E}[S_{40}] &= 10(40) = 400 \text{ minutes}, & \text{Var}(S_{40}) &= 9(40) = 360 \text{ min}^2, \\ \text{SD}(S_{40}) &= \sqrt{360} = 6\sqrt{10} \approx 18.97.\end{aligned}$$

Standardizing 420,

$$z = \frac{420 - 400}{6\sqrt{10}} = \frac{20}{6\sqrt{10}} = \frac{10}{3\sqrt{10}} = \frac{\sqrt{10}}{3} \approx 1.054.$$

The event is “more than,” so it is the upper tail:

$$\begin{aligned}\mathbb{P}(S_{40} > 420) &\approx \mathbb{P}\left(Z > \frac{\sqrt{10}}{3}\right) = 1 - \Phi\left(\frac{\sqrt{10}}{3}\right) \\ &= 1 - 0.8541 = \boxed{0.1459}.\end{aligned}$$

(b) At least 25 books processed in the first 240 minutes. The normal approximation applies to a sum of processing times, not to a count of books, so the event must first be rewritten as a statement about S_{25} . At least 25 books are finished by minute 240 precisely when the 25th book is finished by minute 240, that is when $S_{25} \leq 240$. Note the reversal: *more* books completed corresponds to *less* time elapsed, so the inequality runs the other way. Here

$$\mathbb{E}[S_{25}] = 10(25) = 250 \text{ minutes}, \quad \text{Var}(S_{25}) = 9(25) = 225 \text{ min}^2, \quad \text{SD}(S_{25}) = 15,$$

and standardizing 240 gives

$$z = \frac{240 - 250}{15} = -\frac{10}{15} = -\frac{2}{3} \approx -0.667.$$

The event is “at most,” so it is the lower tail:

$$\begin{aligned} \mathbb{P}(\text{at least 25 books by minute 240}) &= \mathbb{P}(S_{25} \leq 240) \\ &\approx \mathbb{P}(Z \leq -\tfrac{2}{3}) = \Phi(-\tfrac{2}{3}) = \boxed{0.2525}. \end{aligned}$$